

Notes: Cong, Gao, Ponticelli, and Yang (RFS, 2019)

Credit Allocation Under Economic Stimulus: Evidence from China

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1 Overview

This paper studies how a large, policy-driven credit expansion affects *who gets credit* and *what happens after credit is reallocated*. The setting is China’s 2009–2010 stimulus plan, which combined (i) fiscal spending (with a large local-government component) and (ii) a set of monetary/credit policies designed to expand bank lending (lower reserve requirements and lower benchmark lending rates). Using confidential bank–firm loan-level data from the 19 largest Chinese banks matched to manufacturing-firm data, the authors construct a firm-specific exposure to bank credit supply changes and use it to identify causal effects of credit supply on firm borrowing and real outcomes. The central empirical result is that the stimulus-driven credit expansion *reversed* the pre-2008 trend of reallocating credit toward *more productive private firms*: during the stimulus years, credit expansion disproportionately favored (a) state-owned enterprises (SOEs) and (b) firms with a lower average product of capital (a proxy for lower productivity/marginal return to capital). This shift is associated with an increase in credit misallocation and a worsening of ex post loan performance, consistent with implicit government guarantees becoming more salient in recessions.

2 Why this paper matters for productivity and misallocation

2.1 Allocation and the average product of capital

A key productivity lens is that efficient capital allocation tends to push resources toward firms with higher marginal returns to capital. In many firm-level applications, the *average product of capital* (APK) is used as an observable proxy for capital returns:

$$APK_{i,t} \equiv \frac{\text{Value Added}_{i,t}}{K_{i,t}}.$$

If capital is efficiently allocated (subject to measurement and adjustment frictions), firms with persistently high *APK* should expand faster when credit supply is abundant, while low-*APK* firms should shrink or at least not disproportionately absorb the marginal credit. The paper uses *heterogeneity in borrowing responses with respect to APK and state ownership* to diagnose whether the marginal credit is flowing toward higher-return uses or toward politically connected / implicitly guaranteed uses.

2.2 Connection to “TFP/misallocation” logic

Although the paper is framed around credit allocation (rather than a full Olley–Pakes-style or Hsieh–Klenow-style TFP decomposition), the empirical objects map naturally into misallocation measures:

- If stimulus credit expands borrowing more for low-*APK* firms, dispersion in returns to capital can widen and aggregate productivity can fall relative to a benchmark allocation.
- A shift in credit toward SOEs is a *between-group* reallocation (private $\rightarrow$ state), while a shift toward low-*APK* among private firms is a *within-group* reallocation. The paper quantifies

that the between effect is dominant (roughly 70% of the increase in misallocation), with the within-private effect accounting for the remaining 30%.

3 Institutional background: what the stimulus changes

3.1 Two components of the stimulus

The stimulus plan has:

1. **Fiscal plan:** 4 trillion RMB in spending over two years (about 12.6% of 2008 GDP), largely infrastructure and social policies, with local governments financing a large share using LGFVs.
2. **Credit plan:** monetary/financial policies aimed at expanding bank lending to the real economy (lower reserve requirements and benchmark lending rates).

Figures 1–6 in the paper document how aggregate financing and bank lending surged during 2009–2010 and how policy instruments (reserve ratios and benchmark rates) were adjusted.

4 Data construction

4.1 Loan-level micro data

The core micro dataset is confidential regulatory loan data from the China Banking Regulatory Commission (CBRC) covering the 19 largest Chinese banks. Key objects:

- **Bank–firm lending relationships:** loan balances and flows at the bank–firm level.
- **Time dimension:** years spanning pre-stimulus, stimulus (2009–2010), and post-stimulus periods.

4.2 Firm-level manufacturing data

Firms are matched to manufacturing survey data (Annual Survey of Industrial Firms). This provides:

- capital stock K , employment L , sales/value-added, exports, age, public listing indicator,
- state ownership share $\text{StateShare}_{i,t}$,
- productivity proxy $APK_{i,t}$.

5 Identification strategy and econometric models

The key identification problem is to separate *credit supply* from *credit demand*. Firms may borrow more because they have better investment opportunities; the paper instead seeks quasi-exogenous variation in credit supply induced by the stimulus.

5.1 Step 1: bank-level credit supply shocks

Let $L_{b-cj,t}$ be bank b 's *aggregate* loan balance at time t to all borrowers excluding those in firm i 's city c and sector j . Define the bank-level credit growth measure:

$$\Delta \log L_{b-cj,t}.$$

Excluding firm i 's city and sector is designed to net out local/industry demand shocks that could be correlated with firm outcomes.

5.2 Step 2: firm exposure to credit supply (shift-share)

Let $\omega_{bi,t=0}$ be the share of firm i 's borrowing from bank b in a pre-period $t = 0$ (pre-stimulus), and let O_i be the set of firm i 's lenders in $t = 0$. The firm-level exposure measure is:

$$\Delta \tilde{L}_{icjt} \equiv \sum_{b \in O_i} \omega_{bi,t=0} \Delta \log L_{b-cj,t}. \quad (1)$$

Interpretation: $\Delta \tilde{L}_{icjt}$ is large when firm i is connected (pre-stimulus) to banks that expand lending more during the stimulus, *using only those banks' lending to other cities/sectors to purge local demand*. This is the paper's main credit-supply shifter at the firm level.

5.3 Validation: within-firm across-lender test (Khwaja–Mian style)

To validate that $\Delta \log L_{b-cj,t}$ captures bank supply, the paper estimates a bank–firm regression that includes firm $\times$ year fixed effects:

$$\Delta \log loan_{ibcjt} = \alpha + \alpha_{it} + \beta \Delta \log L_{b-cj,t} + \varepsilon_{ibt}, \quad (2)$$

where $\Delta \log loan_{ibcjt}$ is the growth in lending from bank b to firm i . **Logic:** α_{it} absorbs all firm-level demand/investment shocks in year t . Identification comes from comparing, within the same firm-year, how different lenders adjust lending depending on their aggregate supply growth. A positive β supports the credit-supply interpretation.

5.4 Baseline firm-level outcome regression

For a firm-level outcome y_{icjt} (borrowing, investment, employment), the baseline reduced-form specification is:

$$\Delta \log y_{icjt} = \alpha_c + \alpha_j + \alpha_t + \beta \Delta \tilde{L}_{icjt} + \gamma X_{i,t-1} + \varepsilon_{icjt}, \quad (3)$$

where $X_{i,t-1}$ includes lagged firm characteristics (size, exporter, age, public listing).

5.5 Heterogeneity: productivity and state ownership

To study whether marginal credit flows to high-return or politically connected firms, the paper estimates:

$$\Delta \log y_{icjt} = \alpha_c + \alpha_j + \alpha_t + \beta_1 \Delta \tilde{L}_{icjt} + \beta_2 C_{i,t=0} + \beta_3 (\Delta \tilde{L}_{icjt} \times C_{i,t=0}) + \gamma X_{i,t-1} + \varepsilon_{icjt}, \quad (4)$$

where $C_{i,t=0}$ is either $\log APK_{i,t=0}$ or $\text{StateShare}_{i,t=0}$.

5.6 Dynamics: pre vs stimulus vs post

To study changes in the allocation regime over time, the paper estimates a dynamic interaction model:

$$\begin{aligned} \Delta \log y_{icjt} = & \alpha_c + \alpha_j + \alpha_t + \beta_1 \Delta \tilde{L}_{icjt} + \beta_2 (\Delta \tilde{L}_{icjt} \times C_{i,t=0}) + \beta_3 (\Delta \tilde{L}_{icjt} \times \text{Stimulus}_t) \\ & + \beta_4 (\Delta \tilde{L}_{icjt} \times C_{i,t=0} \times \text{Stimulus}_t) + \beta_5 (\Delta \tilde{L}_{icjt} \times \text{Post}_t) + \beta_6 (\Delta \tilde{L}_{icjt} \times C_{i,t=0} \times \text{Post}_t) \\ & + \gamma X_{i,t-1} + \varepsilon_{icjt}. \end{aligned} \quad (5)$$

Here Stimulus_t indicates 2009–2010, and Post_t indicates post-stimulus years (relative to pre-stimulus baseline). **Key object:** the sign of β_2 (pre-stimulus allocation) compared with $\beta_2 + \beta_4$ (stimulus allocation) tells whether allocation shifts toward high- APK (efficient) or toward low- APK (misallocation) during the stimulus.

6 Reading the figures

6.1 Figure 1: Aggregate financing to the real economy

Figure 1 plots Total Social Financing components. The key visual fact is a sharp spike in bank loans during 2009–2010, consistent with a large policy-driven credit expansion. This motivates treating 2009–2010 as a regime shift in credit supply.

6.2 Figure 2: Structure of the stimulus plan

Figure 2 organizes the stimulus into fiscal channels (central and local spending, LGFVs) and credit channels (bank-lending expansion). The figure is mainly institutional: it clarifies that bank credit expansion is a central policy tool, not merely an endogenous outcome.

6.3 Figure 3: Monetary and regulatory policy shifts

Figure 3 documents declines in required reserve ratios and benchmark lending rates around the crisis/stimulus period. The figure supports the interpretation that banks faced policy-induced relaxations of balance-sheet constraints.

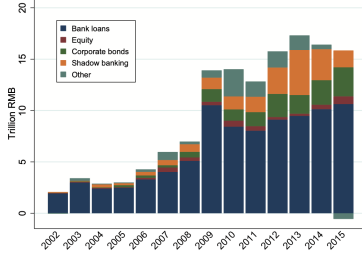


Figure 1
Aggregate financing to the real economy
The category “shadow banking” includes loans by trust companies (trust loans) and entrusted firm-to-firm loans (entrusted loans). The category “other” includes bankers’ acceptances and credit operations categorized under “other” in the TSF data.
Source: Total Social Financing Data Set (TSF) of the People’s Bank of China.

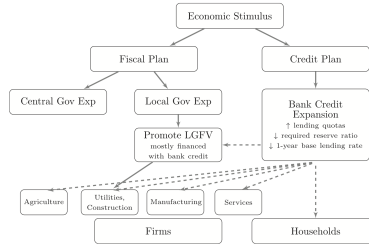


Figure 2
Structure of China economic stimulus plan

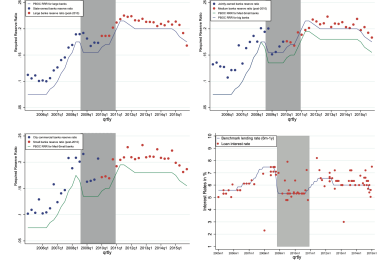


Figure 3
Changes in banking regulation during stimulus years: bank required reserve ratio (RRR) and benchmark lending rate

6.4 Figure 4: Bank reserve ratios and credit growth

Figure 4 is a cross-bank scatter that links pre-stimulus reserve ratios to credit growth during the stimulus. The positive slope suggests that banks with more “room” or more policy-driven reserve changes expanded credit more, generating cross-bank heterogeneity that feeds the paper’s identification.

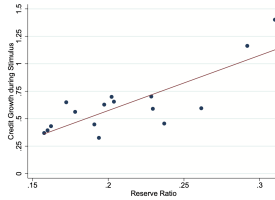


Figure 4
Reserve ratio and credit growth
Reserve ratio at the bank level refers to the year 2007, and it is available from 16 out of 19 banks in our sample. Credit growth is computed as the percentage increase in bank lending between the pre-stimulus years (2007 and 2008) and the stimulus years (2009 and 2010).
Source: banks’ annual reports and China Banking Regulatory Commission.

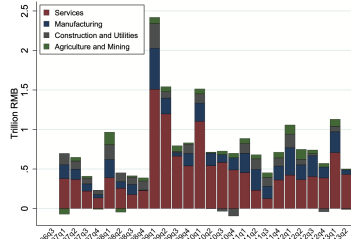


Figure 5
Bank lending to firms by sector: Quarterly data, 2007–2013
To produce this graph we first sum across firms the monetary value of their outstanding loan balance at the end of each quarter. Then we take a quarter-to-quarter difference of the sum.
Source: China Banking Regulatory Commission.

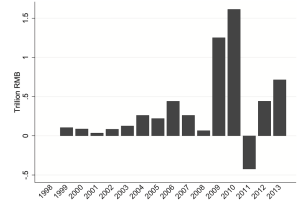


Figure 6
Change in long-term liabilities-manufacturing sector: Yearly data, 1998–2013
To produce this graph we first sum across firms the monetary value of their long-term liabilities at the end of each year. Then we take a year-on-year difference of this sum. To ensure comparability over time, we focus exclusively on manufacturing firms with annual revenues above 20 million RMB (CPI adjusted, in 2000 RMB), for which the survey is effectively a census between 1998 and 2013.
Source: National Bureau of Statistics, Annual Industrial Survey.

6.5 Figure 5: Composition of lending growth by sector

Figure 5 decomposes the surge in firm lending growth across sectors. The broad-based spike in 2009Q1 indicates the credit expansion reached manufacturing and services, not only construction, supporting the use of manufacturing-firm data to study credit allocation.

6.6 Figure 6: Long-term liabilities in manufacturing

Figure 6 shows a dramatic increase in changes in long-term liabilities for manufacturing firms in 2009–2010 relative to surrounding years, consistent with a large expansion in longer-horizon funding during the stimulus.

6.7 Figure 7: Distribution of average product of capital

Figure 7 plots the *APK* distribution separately for private firms and SOEs. A first-order takeaway is that SOEs are shifted toward lower *APK* (lower average return to capital). This makes the paper’s “SOE-favored” credit reallocation naturally interpretable as a move toward lower-return uses of capital.

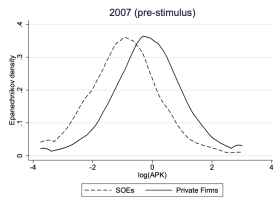


Figure 7
Average product of capital distribution for private firms and SOEs
Source: National Bureau of Statistics, Annual Industrial Survey.

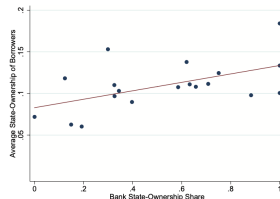


Figure 8
Bank and firm state-ownership connection
Bank State-Ownership Share is the sum of the ownership share of financial institutions under the direct control of the central government, funds under the control of local governments, and state-owned firms. Average State-Ownership of Borrowers is the value-weighted share of a bank lending portfolio to manufacturing firms allocated to SOEs. Both variables refer to the year 2008.
Source: Banks' Annual Reports, China Banking Regulatory Commission and Manufacturing Survey.

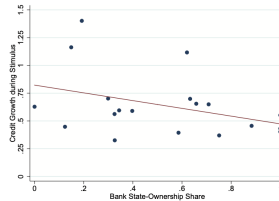


Figure 9
Bank state-ownership and credit growth during stimulus
Bank State-Ownership Share is the sum of the ownership share of financial institutions under the direct control of the central government, funds under the control of local governments, and state-owned firms; ownership data refers to the year 2008. Credit growth is computed as the percentage increase in bank lending between the pre-stimulus years (2007 and 2008) and the stimulus years (2009 and 2010).
Source: Banks' Annual Reports and China Banking Regulatory Commission.

6.8 Figures 8–9: Bank ownership and lending patterns

Figure 8 shows that banks with higher state ownership lend more (in portfolio shares) to SOEs. Figure 9 shows that bank state ownership is *not* positively correlated with stimulus-period credit growth (the slope is negative and marginally significant). Together, these figures suggest:

- *Yes*: state-owned banks are more SOE-oriented in their borrower composition.
- *No*: state-owned banks are not the main drivers of the overall credit expansion.

Hence, “state-owned bank expansion” alone is unlikely to explain the observed economy-wide reallocation toward SOEs; the paper instead emphasizes implicit guarantees in recessions as a broader mechanism.

7 Reading the tables (regressions, coefficients, and economic meaning)

Throughout, standard errors are clustered as in the paper and significance is reported with stars.

7.1 Table 1: Summary statistics

Table 1 provides summary statistics for (i) the full loan dataset and (ii) the matched loan–firm sample used for the main analysis. The matched sample is smaller but has rich firm covariates (including APK and state ownership). The key point is that the matched sample still contains a large number of bank–firm observations with meaningful cross-sectional variation in productivity and ownership.

7.2 Table 2: Persistence of bank–firm credit relationships

Table 2 estimates linear probability models where the outcome is whether a firm receives a new loan from a given lender in year t . The coefficient on the indicator for an existing prior relationship is about 0.95, meaning that conditional on firm×year effects (and also on lender fixed effects), firms overwhelmingly borrow again from their preexisting lenders. This persistence justifies using pre-stimulus lender shares $\omega_{bi,t=0}$ in the exposure measure (1): relationships are stable enough that “who my lenders are” is predetermined.

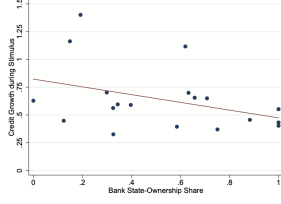


Figure 9
Bank state-ownership and credit growth during stimulus
Bank State-Ownership Share is the sum of the ownership share of financial institutions under the direct control of the central government, funds under the control of local governments, and state-owned firms; ownership data refers to the year 2008. Credit growth is computed as the percentage increase in bank lending between the pre-stimulus years (2007 and 2008) and the stimulus years (2009 and 2010).
Source: Banks' Annual Reports and China Banking Regulatory Commission.

Table 2
Persistence of bank-firm credit relationships

outcome: sample:	New loan from lender _{it,t}	
	all years (1)	stimulus years (2)
preexisting banking relationship	0.949 [0.001]***	0.941 [0.001]***
Year FE	✓	✓
Lender FE	✓	✓
Industry FE	✓	✓
City FE	✓	✓
R ²	0.807	0.789
Observations	882,580	252,167

The outcome variable is a dummy equal to 1 if firm i takes a new loan from bank b at time t . Each observation in the data set is a potential bank-firm relationship, that is, for each firm and year, there is an observation for each potential lender. The independent variable is a dummy equal to 1 if firm i had a preexisting credit relationship with bank b at time $t-1$. Standard errors clustered by firm. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 3
Bank credit supply and loan

outcome: sample:	$\Delta \log loan_{it}$					
	all years 2006–2013			stimulus years 2009–2010		
	all firms	multi-lender	all firms	all firms	multi-lender	all firms
$\Delta \log loan_{it}$	0.140 [0.000]***	0.146 [0.000]***	0.128 [0.000]***	0.165 [0.000]***	0.152 [0.000]***	0.154 [0.000]***
Year FE	✓	✓	✓	✓	✓	✓
Industry FE	✓	✓	✓	✓	✓	✓
City FE	✓	✓	✓	✓	✓	✓
Pre(Nonstimulus)	✓	✓	✓	✓	✓	✓
Pre × Year FE	✓	✓	✓	✓	✓	✓
R ²	0.052	0.052	0.052	0.056	0.056	0.056
Observations	177,289	177,289	143,125	80,908	80,908	11,225

The unit of observation is a bank-firm credit relationship. The dependent variable is yearly change in the log of the outstanding loan balance from bank b to firm i . Firm observations are from the 100 largest firms in each of the 100 cities. The independent variable is yearly change in the log of the outstanding loan balance from bank b to firm i . Firm observations are from the 100 largest firms in each of the 100 cities. The dependent variable is yearly change in the log of the outstanding loan balance from bank b to firm i . Firm observations are from the 100 largest firms in each of the 100 cities. The dependent variable is yearly change in the log of the outstanding loan balance from bank b to firm i . Firm observations are from the 100 largest firms in each of the 100 cities. Standard errors are clustered at the bank-firm level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

7.3 Table 3: Validating the bank supply measure (Equation (2))

Table 3 estimates (2). The coefficient β is positive and significant across specifications (roughly 0.14–0.17). Interpretation:

- Within the same firm-year, lenders that expand aggregate lending more (outside the firm's city/sector) expand lending to the firm more.
- This is hard to reconcile with pure firm demand shocks, because firm-year fixed effects absorb them.

Hence, $\Delta \log L_{b-cj,t}$ behaves like a bank-level supply shifter.

7.4 Table 4: Average effects of credit supply during the stimulus

Table 4 estimates (3) in stimulus years (2009–2010) with outcomes: (i) firm borrowing growth, (ii) investment rate (change in fixed assets over lagged assets), and (iii) employment growth. Key coefficients (column 1 / 2):

- **Borrowing:** $\beta \approx 1.01$ (or 1.00 with controls). A 1% higher exposure to bank credit supply translates into about a 1% higher firm borrowing growth.
- **Investment:** $\beta \approx 0.078$ (or 0.070). Credit supply increases raise investment intensity.
- **Employment:** $\beta \approx 0.346$ (or 0.297). Credit supply increases raise employment.

Economic meaning: the stimulus credit expansion had real effects (investment and employment), not just balance-sheet reshuffling.

7.5 Table 5: Heterogeneous effects (misallocation diagnosis)

Table 5 estimates (4) with borrowing growth as the outcome.

Productivity heterogeneity (columns 1–4). The interaction $\Delta \tilde{L}_{icjt} \times \log APK_{i,t=0}$ is *negative* (about -0.059 with controls; also negative in the private-only sample). This implies that stimulus-driven credit supply increases raise borrowing *more* for *lower-APK* firms.

$$\frac{\partial \Delta \log loan_{it}}{\partial \Delta \tilde{L}_{it}} = \beta_1 + \beta_3 \log APK_{i,t=0}, \quad \beta_3 < 0.$$

So the marginal credit is tilted toward lower average returns to capital, a direct misallocation signature.

Table 4
The effect of bank credit supply on firm-level outcomes: Loans, investment, and employment: Stimulus years (2009–2010)

outcome:	$\Delta \log loan_{it}$		$\frac{\Delta K_{it}}{\Delta_{it-1}}$		$\Delta \log Employment_{it}$	
	(1)	(2)	(3)	(4)	(5)	(6)
Δ_{it-1}	1.010	1.003	0.078	0.070	0.346	0.297
	[0.086]***	[0.086]***	[0.032]**	[0.033]**	[0.102]**	[0.100]***
Year FE	y	y	y	y	y	y
Industry FE	y	y	y	y	y	y
City FE	y	y	y	y	y	y
Firm Characteristics	y	y	y	y	y	y
R^2	0.002	0.004	0.254	0.258	0.227	0.247
Observations	11,068	11,068	11,068	11,068	11,068	11,068

The unit of observation is a firm. The dependent variables are: the yearly change in the log of total outstanding bank loan balance in Columns (1) and (2), the yearly change in physical capital scaled by lagged total assets in Columns (3) and (4), the yearly change in the log of average number of workers in Columns (5) and (6). Firm characteristics are: firm size in terms of number of workers (in logs), export status (dummy equal to 1 if a firm has positive value of export in a given year), firm age, and a dummy equal to 1 if the firm is publicly traded. Firm characteristics are observed in year $t-1$. Standard errors are clustered at the city level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 5
Heterogeneous effects of bank credit supply: Stimulus years (2009–2010)

outcome:	$\Delta \log loan_{it}$				
sample:	all firms		StateShare $_{i,t=0} > 0$		all firms
	(1)	(2)	(3)	(4)	(5)
Δ_{it-1}	0.993	0.985	0.961	1.245	0.974
	[0.088]***	[0.089]***	[0.092]***	[0.257]***	[0.085]***
$\log APK_{i,t=0}$	0.047	0.047	0.052	-0.002	0.054
	[0.008]***	[0.008]***	[0.008]***	[0.021]	[0.005]***
$\Delta_{it-1} \times \log APK_{i,t=0}$	-0.059	-0.060	-0.059	0.037	
	[0.027]**	[0.027]**	[0.028]**	[0.079]	
$\Delta_{it-1} \times \text{StateShare}_{i,t=0}$				0.376	
				[0.119]**	
$\Delta_{it-1} \times \text{StateShare}_{i,t=0} \times I(\text{stimulus})$				-0.074	
				[0.027]**	
Year FE	y	y	y	y	y
Industry FE	y	y	y	y	y
City FE	y	y	y	y	y
Firm Characteristics	y	y	y	y	y
R^2	0.007	0.009	0.101	0.220	0.100
Observations	11,068	11,068	9,254	1,787	11,068

The unit of observation is a firm. The dependent variable is the yearly change in the log of total outstanding bank loan balance. Firm characteristics are: firm size in terms of number of workers (in logs), export status (dummy equal to 1 if a firm has positive value of export in a given year), firm age, and a dummy equal to 1 if the firm is publicly traded. Firm characteristics are observed in year $t-1$. Standard errors are clustered at the city level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 6
Dynamic of credit allocation across firms: All years (2006–2013)

outcome:	$\Delta \log loan_{it}$					
sample:	all firms			StateShare $_{i,t=0}$		
	(1)	(2)	(3)	=0	>0	
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{it-1} \times \log APK_{i,t=0}$	0.098	0.093	0.117	0.100	-0.022	
	[0.045]**	[0.045]**	[0.050]**	[0.053]**	[0.118]	
$\Delta_{it-1} \times \log APK_{i,t=0} \times I(\text{stimulus})$	-0.156	-0.152	-0.152	-0.166	0.041	
	[0.047]***	[0.047]***	[0.060]**	[0.057]***	[0.142]	
$\Delta_{it-1} \times \log APK_{i,t=0} \times I(\text{post})$	-0.067	-0.064	-0.020	-0.067	0.195	
	[0.060]	[0.061]	[0.067]	[0.070]	[0.171]	
$\Delta_{it-1} \times \text{StateShare}_{i,t=0}$					-0.465	
					[0.211]**	
$\Delta_{it-1} \times \text{StateShare}_{i,t=0} \times I(\text{stimulus})$					0.855	
					[0.240]***	
$\Delta_{it-1} \times \text{StateShare}_{i,t=0} \times I(\text{post})$					0.672	
					[0.202]**	
Δ_{it-1}	1.308	1.296	1.432	1.308	1.166	1.311
	[0.098]***	[0.099]***	[0.122]**	[0.101]**	[0.351]**	[0.096]***
$\Delta_{it-1} \times I(\text{stimulus})$	-0.325	-0.319	-0.338	-0.365	0.090	-0.348
	[0.133]**	[0.135]**	[0.154]**	[0.142]**	[0.380]	[0.134]***
$\Delta_{it-1} \times I(\text{post})$	0.210	0.208	-0.249	0.137	1.034	0.182
	[0.149]	[0.149]	[0.210]	[0.155]	[0.488]**	[0.147]
$\text{StateShare}_{i,t=0}$					0.078	
					[0.040]*	
$\text{StateShare}_{i,t=0} \times I(\text{stimulus})$					-0.134	
					[0.045]**	
$\text{StateShare}_{i,t=0} \times I(\text{post})$					0.029	
					[0.003]***	
$\log APK_{i,t=0}$	0.013	0.014	-0.001	0.007	0.046	0.029
	[0.009]	[0.009]	[0.012]	[0.010]	[0.023]**	[0.003]***
$\log APK_{i,t=0} \times I(\text{stimulus})$	0.031	0.030	0.031	0.039	-0.036	
	[0.011]**	[0.011]**	[0.014]**	[0.013]**	[0.031]	
$\log APK_{i,t=0} \times I(\text{post})$	0.011	0.010	0.006	0.013	-0.027	
	[0.010]	[0.010]	[0.012]	[0.011]	[0.026]	
Year FE	y	y	y	y	y	y
Industry FE	y	y	y	y	y	y
City FE	y	y	y	y	y	y
Firm Characteristics	y	y	y	y	y	y
Firm FE	-	-	-	-	-	-
R^2	0.068	0.060	0.144	0.070	0.123	0.070
Observation	46,583	46,583	42,938	39,135	7,440	46,583

The unit of observation is a firm. The dependent variable is the yearly change in the log of total outstanding bank loan balance. $I(\text{stimulus})$ is a dummy equal to 1 for the years 2009 and 2010. $I(\text{post})$ is a dummy equal to 1 for the years 2011 to 2013. Data covers the period 2006 to 2013. Firm characteristics are: firm size in terms of number of workers (in logs), export status (dummy equal to 1 if a firm has positive value of export in a given year), firm age, and a dummy equal to 1 if the firm is publicly traded. Firm characteristics are observed in year $t-1$. Standard errors are clustered at the city level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

State-ownership heterogeneity (columns 5–7). The interaction $\Delta \tilde{L}_{icjt} \times \text{StateShare}_{i,t=0}$ is *positive* (about +0.376). This implies stimulus credit expansion raises borrowing more for more state-owned firms. Splitting the sample confirms larger sensitivity for SOEs than for purely private firms. Together with Figure 7 (SOEs have lower APK), this points to a *between-group* reallocation toward lower-return firms.

7.6 Table 6: Dynamics of credit allocation across periods

Table 6 estimates the dynamic model (5).

Productivity channel. In the pre-stimulus period, $\Delta \tilde{L} \times \log APK$ is positive (≈ 0.098): credit supply increases are more associated with borrowing by higher- APK firms (allocation improving). During stimulus years, the triple interaction is negative (≈ -0.156), implying a reversal:

$$(\Delta \tilde{L} \times \log APK) \text{ effect in stimulus} \approx 0.098 - 0.156 \approx -0.058.$$

So the stimulus flips the sign of “credit flows to high- APK ” into “credit flows to low- APK .”

State-ownership channel. Pre-stimulus, $\Delta \tilde{L} \times \text{StateShare}$ is negative (≈ -0.465): supply expansions favor private firms. Stimulus years add +0.855, so the net effect becomes positive:

$$-0.465 + 0.855 \approx 0.390,$$

indicating a strong shift toward SOEs. Post-stimulus remains positive (adding +0.672), suggesting persistence.

7.7 Table 7: Real effects and ex post loan performance over time

Table 7 applies the dynamic framework to investment, employment, and nonperforming loans.

Panel A: Investment. Pre-stimulus, $\Delta \tilde{L} \times \log APK$ is positive (≈ 0.191): high- APK firms invest more when credit expands. Stimulus interaction is negative (≈ -0.255), reversing the pattern (net ≈ -0.064). This shows that the misallocation is not only in borrowing but also in real capital formation.

Panel B: Employment. Pre-stimulus, $\Delta \tilde{L} \times \log APK$ is positive (≈ 0.710), but the stimulus interaction is negative (≈ -0.672), largely flattening differential employment growth across productivity types. The state-share interactions show employment effects tilt toward SOEs during stimulus.

Panel C: Loan performance (NPL). Pre-stimulus, $\Delta \tilde{L} \times \log APK$ is negative (≈ -0.151): lending expansions are associated with *lower* subsequent default for high- APK firms. During stimulus, the interaction becomes $+0.173$, closing the performance gap between high- and low- APK borrowers. This is consistent with credit flowing toward firms with worse fundamentals (or toward firms that are more likely to be implicitly supported), increasing misallocation and weakening the informational content of default outcomes.

8 Appendix A model: a structured mechanism (implicit guarantees in recessions)

The paper cannot directly test implicit guarantees at the micro level, so it provides a model that rationalizes the empirical reversal.

8.1 Environment

Time is discrete. There are two firm types:

- **State-connected (S) firms** (SOEs): lower productivity but enjoy implicit guarantees.
- **Private (P) firms**: higher productivity but face limited pledgeability.

Production:

$$y_{S,t} = k_{S,t}^\alpha (\tilde{A}_{S,t} n_{S,t})^{1-\alpha}, \quad y_{P,t} = k_{P,t}^\alpha (\tilde{A}_{P,t} n_{P,t})^{1-\alpha},$$

with $\tilde{A}_{S,t} = A$ w.p. μ_t and 0 otherwise, and $\tilde{A}_{P,t} = \chi A$ w.p. μ_t and 0 otherwise ($\chi > 1$ implies P firms are more productive).

8.2 Two credit-market frictions

1. Limited pledgeability for P firms:

$$R_{P,t} l_{P,t} \leq \eta \pi_t(k_{P,t}, n_{P,t}), \quad \eta \in (0, 1),$$

so P firms cannot fully pledge cash flows to lenders.

- ### 2. Bailout probability for S firms:
- when S firms fail, the state repays with probability $b > 0$. Lenders therefore expect repayment probability $\mu_t + (1 - \mu_t)b$, implying cheaper funding for S firms. This generates a dual-track interest rate wedge summarized in the paper as

$$R_{S,t} = \delta R_{P,t}, \quad \delta = \frac{\mu_t}{\mu_t + (1 - \mu_t)b} \in (0, 1).$$

8.3 Key equilibrium implication

When the economy enters a recession (lower μ_t), the value of implicit guarantees rises: the wedge between S and P borrowing conditions becomes more important exactly when default risk is higher. If the government simultaneously expands aggregate credit supply Q_t (stimulus), the model implies the additional credit is disproportionately allocated to S firms, crowding out P firms. This produces:

- a reversal in reallocation toward productive private firms,
- a rise in the credit share (and factor shares) of state-connected firms,
- a persistence of misallocation if credit supply remains elevated.

Figure A1 visualizes these dynamics in simulated paths for the share of bank credit going to S firms.

9 Conclusion

9.1 Summary

This paper studies how a large, policy-driven credit expansion reallocates bank credit across firms and how this reallocation translates into real outcomes. Using confidential loan-level data from the nineteen largest Chinese banks matched to manufacturing-firm data, the paper identifies plausibly exogenous firm-level credit-supply exposure during China’s 2009–2010 stimulus. The central conclusion is that the stimulus-driven credit expansion *did not simply increase aggregate lending*: it also **changed the allocation of credit across firms in a way that worsened capital allocation**. In particular, **new credit disproportionately flowed to state-owned/state-controlled firms and to firms with lower initial (average) product of capital**, reversing the pre-2008 pattern in which credit was increasingly reallocated toward more productive private firms. Moreover, the shift toward state-connected borrowers **did not quickly unwind after 2010**, indicating persistent allocation effects beyond the official stimulus window.

9.2 Main empirical conclusions (with key quantitative anchors)

1. **The stimulus credit expansion materially increased firm borrowing and real activity.** During 2009–2010, firms more exposed to stimulus-driven bank credit supply experienced larger increases in: (i) borrowing, (ii) investment, and (iii) employment. In Table 4, the preferred specification implies that a **1%** larger increase in credit supply from pre-stimulus lenders translates into:

- an increase in firm borrowing of **similar magnitude** (loan growth moves about one-for-one with supply),
- about **0.1 percentage points** higher investment (as a share of lagged assets),
- and about **0.3%** higher employment growth.

2. **Credit allocation became more distorted during the stimulus years.** Table 5 shows that, during 2009–2010, the marginal effect of credit supply on borrowing was:

- **stronger for firms with higher state ownership**, and
- **stronger for firms with lower initial capital productivity** (among private firms).

A useful calibration from the text: in response to a one-standard-deviation change in credit supply, **fully state-owned firms** increased borrowing by about **15.7%**, versus **11.3%** for **fully private firms** (i.e., the borrowing response is about **39% larger** for SOEs).

3. **The rise in misallocation has a clear decomposition: mostly “between” SOEs and private firms.** The paper decomposes the stimulus-period worsening in allocation into:

- a **between effect** (credit shifting from private firms to SOEs), and
- a **within-private effect** (credit shifting toward less productive private firms).

The paper’s accounting implies that roughly **70%** of the increase in misallocation is driven by the **between effect** and **30%** by the **within-private effect** (discussion around Table 5).

4. **The pre-2008 reallocation toward productive firms reversed in 2009–2010 and did not fully revert afterward.** Table 6 extends the analysis to 2006–2013 and shows:

- **Pre-stimulus (2006–2008):** credit supply exposure is associated with *more* borrowing for *higher* initial capital productivity firms (consistent with reallocation toward productive firms),
- **Stimulus (2009–2010):** this pattern *reverses* (credit grows relatively more for initially *less* productive firms),
- **Post-stimulus (2011–2013):** the shift toward SOEs remains economically and statistically present; the paper reports that the post-stimulus SOE-shift coefficient is not statistically different from the stimulus-period coefficient, consistent with persistence beyond 2010.

5. **Loan performance patterns are consistent with an increase in implicit guarantees during and after the stimulus.** Panel C of Table 7 shows that the pre-stimulus gap in default rates between high- and low-productivity firms **shrinks starting in 2009**. The SOE-private gap in ex post loan performance also **closes** beginning in the stimulus years. The interpretation advanced in the paper is that **implicit government guarantees** become more important in downturns/stimulus periods, reducing measured default differentials across borrower types through refinancing and/or direct and indirect government support.

9.3 Economic intuition: the mechanism that ties the facts together

The intuition is that the stimulus increases aggregate credit supply, but the *incremental* credit is not neutral across borrowers when the state can implicitly insure politically connected firms. A coherent causal narrative consistent with the paper is:

1. **In normal times, credit reallocation can favor productive private firms.** Before 2008, China exhibited a reallocation process in which more productive (often private) firms expanded faster and obtained more external finance, consistent with credit moving toward higher marginal returns.
2. **In a downturn, the value of state-connectedness rises because guarantees become more salient.** When recession risk rises, governments have stronger incentives to prevent distress among state-connected firms (e.g., for employment and social stability), and banks have stronger incentives to align with these priorities (career concerns, political costs, and expectations of support).
3. **A credit expansion interacting with guarantees shifts marginal lending toward protected borrowers.** When the policy pushes banks to expand lending, the incremental loans are disproportionately directed to borrowers perceived as safer *because they are guaranteed*, not because they have higher productivity. This generates the observed reallocation toward SOEs and (within private firms) toward lower capital productivity borrowers.
4. **Real effects follow the new allocation, and measured “loan quality” gaps compress.** Because protected borrowers receive more credit, they invest more and (especially for SOEs) expand employment more during the stimulus. Meanwhile, refinancing/guarantees reduce observed default differentials, consistent with the closing of ex post performance gaps.

9.4 Why the paper matters beyond China

The paper’s broader takeaway is that in emerging economies with severe financial frictions and politically mediated guarantees, credit stimulus can have **unintended allocation consequences**. Even if aggregate credit expansion stabilizes activity in the short run, it can **reverse efficient reallocation dynamics** and increase misallocation, with potentially persistent effects on productivity growth.

9.5 Takeaway

China's 2009–2010 credit stimulus increased lending and real activity, but it also redirected marginal credit toward state-connected and low-capital-productivity firms, reversing pre-2008 reallocation toward productive private firms and leaving a persistent imprint consistent with countercyclical implicit government guarantees.